

Facilities Planning Group Project (manufacturing 5 products, 4 processes)

Wiring.com is a company that manufactures wire-based products. It has rich knowledge in the steel and wire industry. The company recently experienced a rapid increase in their netting sales. The company was not able to meet the increase in demand at their current facility and therefore require a bigger facility. Due to various reasons, they could not buy the properties adjacent to the current facility. Thus, the company has purchased a *new* site for development within a 10 km range of their current facility. The site is in the Sedibeng District in Gauteng, South Africa, with a notable size of 50 HA. Currently, the proposed new site has no structures or buildings, and the new manufacturing plant needs to be built from the conceptual design phase. As part of team, management expects a detailed layout within the next 2 months for their consideration.

Operation specifications

The proposed new site will procure (buy) Steel coils (2,5 ton each) as a *raw material* and will perform the following operations:

- **Operation A:** Drawing the rods to the correct wire diameter
- **Operation B:** Galvanising of wire
- **Operation C:** Producing various sizes of netting with specialized netting machines
- **Operation D:** Packaging and packing of finished goods

Research operational equipment used to make netting for better understanding of the processes. Management expects you to recommend equipment as part of the production process design. It will also help to present a process flow chart.

Product specifications

The following products of netting are manufactured:

- **Product A:** Bird Netting
- **Product B:** Chicken Netting
- **Product C:** Jackal Netting
- **Product D:** Pig Netting
- **Product E:** Welded mesh (industrial fencing)

1. Operation A - Wire drawing of rods to the correct sizes

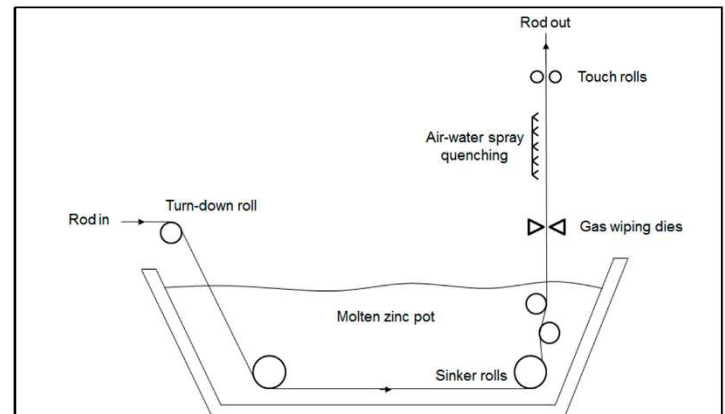
Steel wire coils (5mm diameter and 7000m length) are used as the raw material for the entire operation. The wire must be drawn to the required diameters required for the final products. This drawn wire may then be rolled into coils if required before galvanization.

2. Operation B - Galvanising of wire

workout how many km of wire go through bath before it is empty. Use average surface area and assume thickness of galvanisation

This operation happens in a bath where the wire is drawn through liquid. **Zinc ingots** have to be bought to keep the process going. See below for concept process diagram.

Input Drawn wire to the required wire diameter (sizes 0.6, 0.9, 1.8, 2mm) for specific netting product
Output Galvanized wire in recoiled form WIP of Product A, Product B, Product C, Product D, Product E.



3. Operation C: Producing various sizes of netting

Production rates

Production rates for each operation is based on the demand of the final product (netting). Thus operation A's requirements are based on operation's B requirements which is based on operation C's requirements. These calculations need to be shown as evidence and for management consideration if they want to question specifics. A total of **26 000 tons per annum** of netting products is the *projected customer demand* for Wiring.com. A split between the market demands of the various sizes is provided in the specifications table below.

Product specifications

Product	Height of roll (m)	Aperture (mm)	Wire Diameter (mm)	Roll Length (m)	Weight (kg) per roll	Galvanized finished (Light or Heavy)	Customer requirements tons per annum
Bird Netting	0.9	13	0.71	50	22.5	Light	4000 / year
Chicken Netting	1.8	25	0.9	50	41.9	Light	5500 / year
Jackal Netting	0.6	75	1.8	50	19.5	Light	2500 / year
Pig Netting	0.9	90	1.8	30	14.7	Light	8000 / year
Stainless steel welded mesh	2	50	2.5	30	70	Heavy	6000 / year

Wat is die dikte van 'n rol?

							26 000 tons / yr
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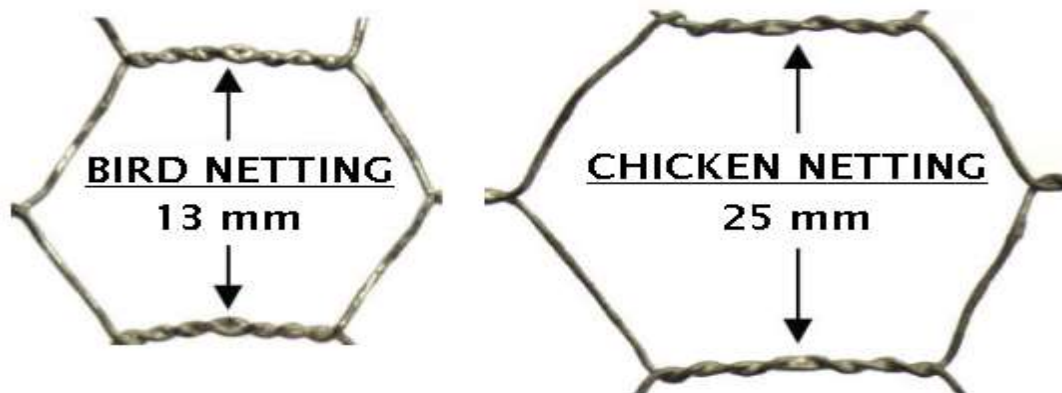


Figure-1: Example of 'aperture' sizes of each netting product.

4. Operation D: Packaging for delivery and sales



You can make your own decision on how to PACKAGE the netting most effectively. Justify the decision based on research done. The packaging process has to be part of the facility design.

Figure 2 Images of Rolls of netting packaged in pallets for sales.

(source: <https://images.app.goo.gl/Rxtd6UQQ3VNLxYK66>)

Management policies to be considered when designing facilities:

- **Parking** must be provided for all personnel. Personnel include management, administration and operators. Show all calculations on determining parking space.
- **Lockers** for plant personnel are provided as well as a *cafeteria* providing basic food, but no Health services are provided. There should be enough Rest Rooms provided in both offices as well as in the Manufacturing areas, as per standards.

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- *Offices* are mainly for plant **personnel** including Plant management, Operations supervisors, Sales, Purchasing, Finance and Inventory management departments, Administration personnel, Engineering, Operations, Maintenance and Spare parts storage rooms for maintenance. Also allow room for Reception offices, to receive customers and suppliers bringing the paperwork for deliveries.
- Assume that **customers can pick up small orders** themselves within a week of being made, but large orders that can fit onto a truck will be delivered to customers. [you may define small and large orders yourself, and you may choose truck sizes to help define large orders]. This implies that the Finished Goods store will not have to be that large, but it must still be able to store 2 days product before dispatch.
- Make logical assumptions regarding Facility support systems eg. Lightning, Atmospheric, Safety and other factory requirements.
- Management is considering sustainable energy options. Recommend at least 2 strategies they can consider incorporating into the new facility design.

Assumptions and Constraints given by the customer

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- Assume 220 working days per annum/year. **9hr day and 1hr lunch. 8hrs working times.**
- Assume a 4% scrap rate in the first operation. (scrap happens during the first few processes especially so there should be storage space allowed for it to be placed before it can be taken away from site)
- 1 ton = 1000kg
- Assume galvanization has no marginal effect on mass.
- Raw material stock must be for 3 days production demand. Raw materials must be kept near operations, but safely and with easy access for the supplier to deliver weekly.
- The company will not necessarily use all 50Ha for the operations, but only **75% may be used** for buildings, production space and storage. 25 % must be allocated to roads (deliveries and dispatch), parking and green buffer zones.
- Inside the facility 10% expansion capacity must be considered in the design as the company expect growth within the next 5-10years.